

Chapter 19.1

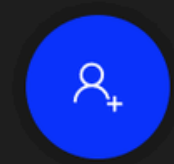
Hierarchical Clustering

தமிழில்

Less than 1 Hour

AI COACH JOHN

CEO & FOUNDER



CLUSTERING

"When there is no specific o/p $\rightarrow$ i.e, No 'y' "



①

'Unsupervised'

Grouping // Clustering
"Similar items get into 1 group".
 ξ
"Each & every group is different from one another."



color	shape	size	label
Green	Round	small	Grape
Yellow	Round	Small	Lemon
Red	Round	Medium	Apple
Orange	Round	Medium	Orange

Kid Training.

'Supervised'

Orange | Round | Medium | ? // Testing

'Evaluation':



Use Case:

"ANUSHTA" $\Rightarrow$ 'Restaurant Owner' $\Rightarrow$ 100K $\Rightarrow$ '10 years'.

(100K)

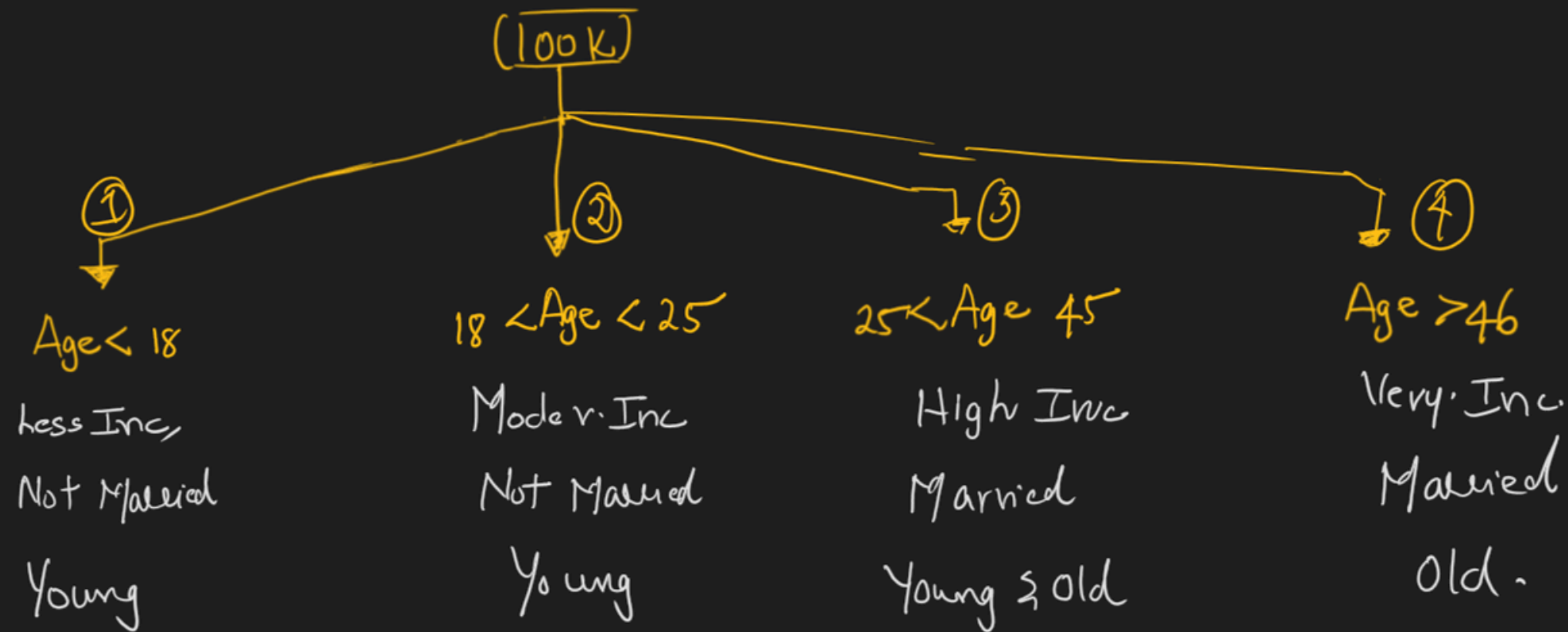
"similar items get into 1 group".

"Each ξ every group is different from one another."

Orange | Round | Medium | : // testing

Use Case:

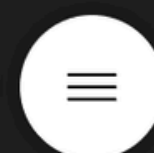
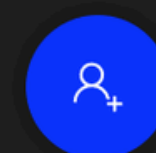
'ANUSHA' $\Rightarrow$ 'Restaurant Owner' $\Rightarrow$ 100 K $\Rightarrow$ '10 years'.



BPO Customer Care Executive Experience

- 1. Silver Customers**
- 2. Gold Customers**
- 3. Titanium Customers**
- 4. Platinum Customers**

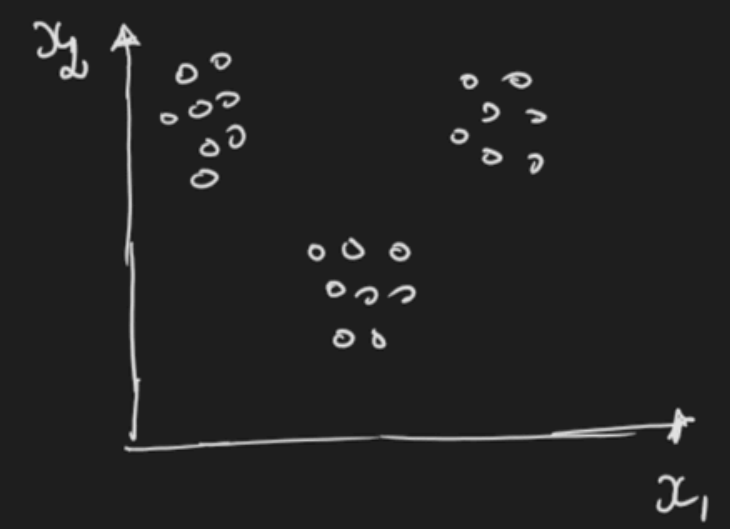




3 Types of Clustering:

- ① Hierarchical Clust. (Agglomerative Clust) → Bottom Up Approach.
- ② K-Means Clust (Divisive Clust) → Top Down Approach.
- ③ DBSCAN Clust.

1) Hierarchical Clustering:



TA: For 2-D, humans can plot with a graph & easily form the group.
But for higher dimensions, humans cannot visualize the groups.

// That's why we rely on
MACHINES for clustering //

4.1 Make 5 clusters.

You are screen sharing

Stop Share

Graphy


$$\{1\} \{2\} \{3\} \{4\} \{5\}$$
$$\{1\} \quad \{2\} \quad \{3, 4\} \quad \{5\}$$
$$\{1, 2\} \quad \{3, 4\} \quad \{5\}$$
$$\{1, 2\} \quad \{3, 4, 5\}$$
$$\{1, 2, 3, 4, 5\}$$

with the help
of DENDROGRAM



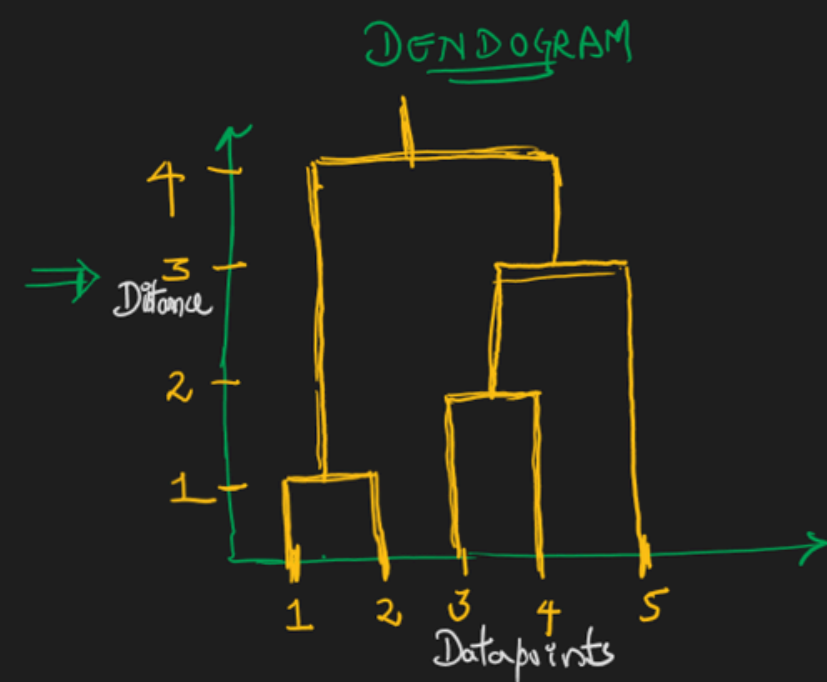
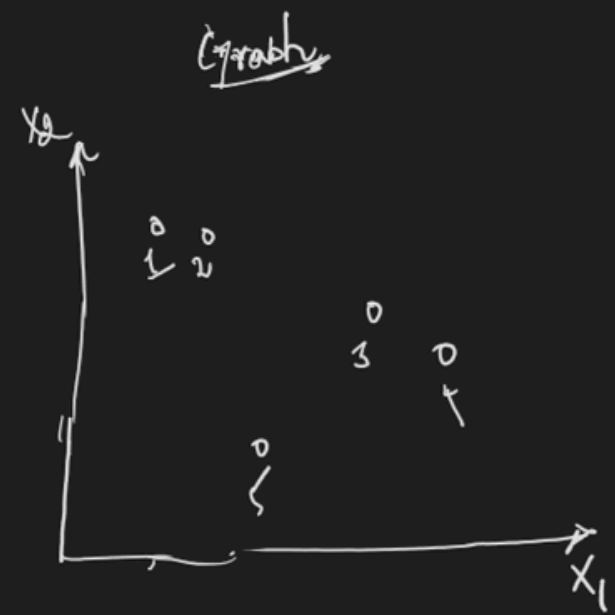
② Combine 1 record

" linkage

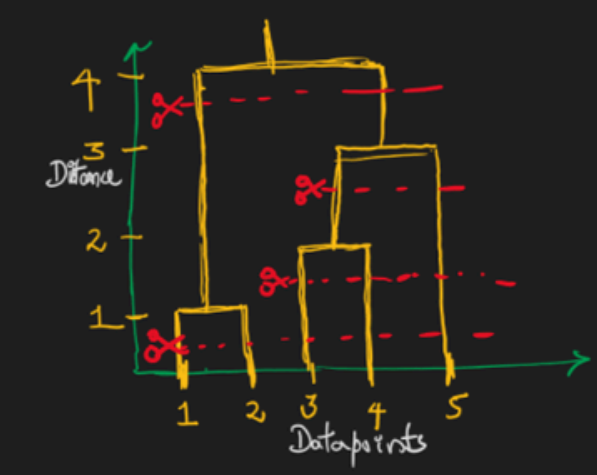




with the help
of DENDROGRAM

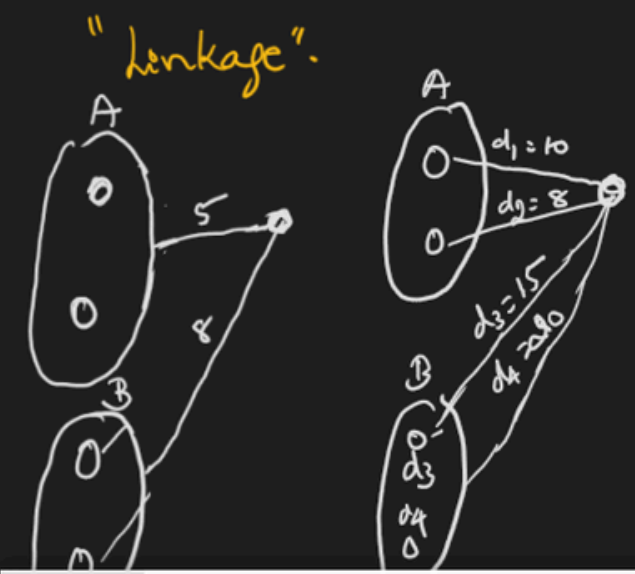


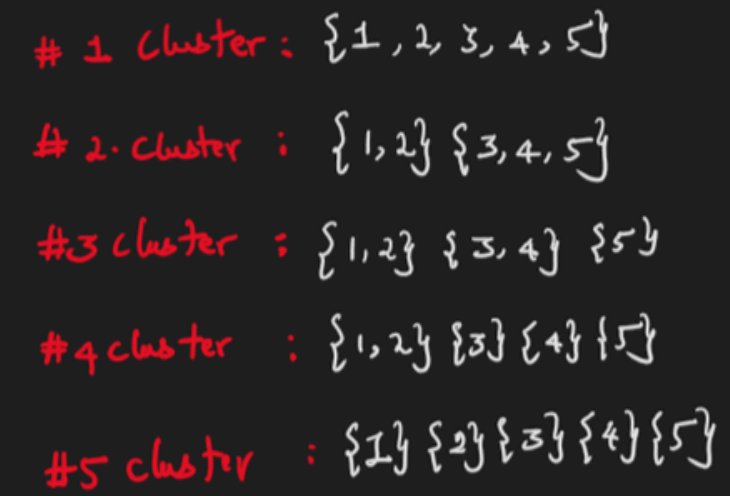
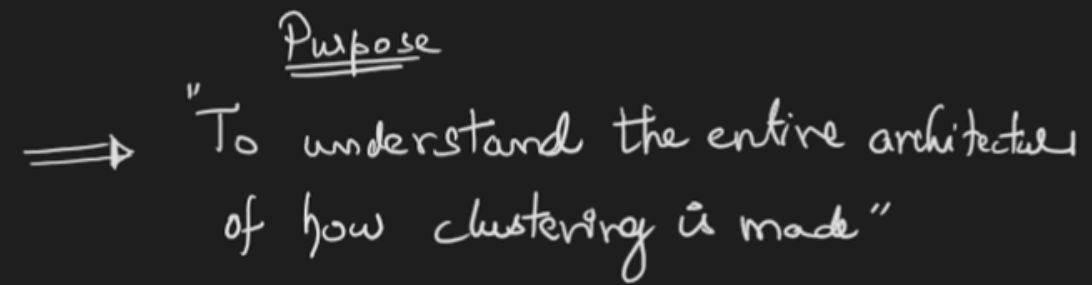
Purpose
⇒ "To understand the entire architecture of how clustering is made"



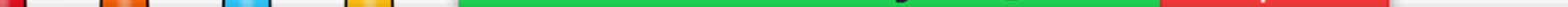
- ① Find the distance b/w each point?
- Euclidean distance (L_2) = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ ✓
 - Manhattan distance (L_1) = $|x_2 - x_1| + |y_2 - y_1|$

- ② Combine 1 record to a cluster?
- Max(d_1, d_2) ⇒ complete linkage ⇒ Max Distance
 - Min(d_1, d_2) ⇒ Single linkage ⇒ Minimum Distance
 - Avg(d_1, d_2) ⇒ Average linkage ⇒





- $\text{Max}(d_1, d_2) \Rightarrow$ Complete linkage $\rightarrow$ Max distance
- $\text{Min}(d_1, d_2) \Rightarrow$ Single linkage $\rightarrow$ Minimum distance
- $\text{Avg}(d_1, d_2) \Rightarrow$ Average linkage $\rightarrow$

A horizontal toolbar from a Zoom application. It features several drawing tools (pens and highlighters in various colors) on the left. In the center, a green banner displays the text "You are screen sharing" with a small icon of a person and a screen. To the right of this banner is a red button with a white square icon and the text "Stop Share". Further right are icons for adding a new participant, undo, and redo.



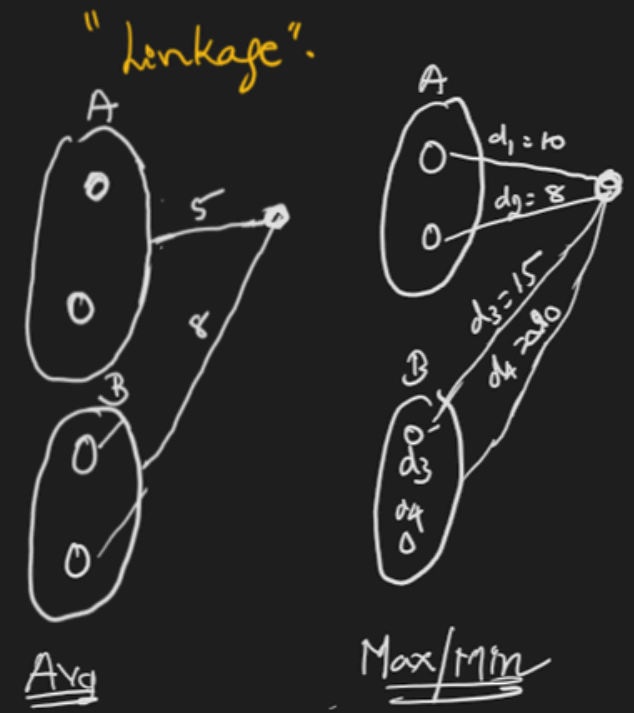
① Find the distance b/w each point?

- Euclidean distance (L2) = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ ✓ 
- Manhattan distance (L1) = $|x_2 - x_1| + |y_2 - y_1|$ 

② Combine 1 record to a cluster?

"Linkage".

- Max(d_1, d_2) $\Rightarrow$ Complete linkage $\Rightarrow$ Max Distance
- Min(d_1, d_2) $\Rightarrow$ Single linkage $\Rightarrow$ Minimum Distance
- Avg(d_1, d_2) $\Rightarrow$ Average linkage $\Rightarrow$



③ Combine cluster with another cluster:

- Single linkage
- Complete linkage
- Average linkage.

② K-Means Cl